TREES AND THEIR DEGREES

MARIE-CAMILLE DELARUE

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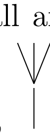
1. Introduction

In the course of our study of the family of Higman–Thompson groups, there came to be a question that was very easy to answer in small dimensions and much harder in higher dimensions. The fundamental reason for this was that it was easy to come up with a reasonable visualization in low dimensions, and much harder in higher dimensions because of the limitations of 3-dimensional space. Therefore, we tried to come up with a good visualization in higher dimensions.

The following document explains this problem and the framework we came up with to answer it in higher dimensions. For extra context on the Thompson groups and their homology, please consult the other file in this folder. Reading the following document is however completely necessary to understanding the specific question we were working on as well as its implementation.

2. Trees

We will consider finite rooted trees, and we call an a-**junction** a vertex of the tree with one parent and a children: For instance,  is a 3-junction.

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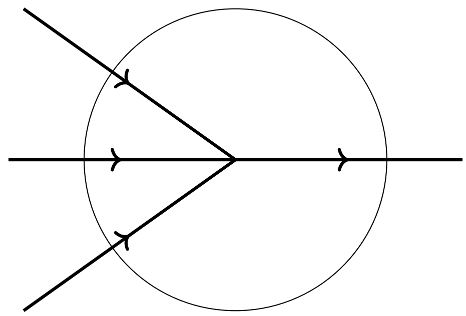
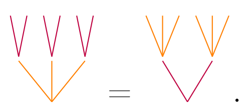
2.1. **Trees with simple junctions.** Let us start by considering the following problem. Consider a finite rooted tree which only has a-junctions, for a an integer. We will think of trees as being oriented from the leaves to the root. Consider the following picture:



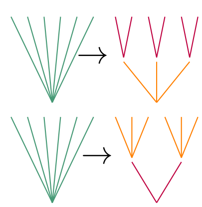
We will call the **degree** of a junction the number of incoming edges minus the number of outgoing edges in a small circle around the junction. For any a-junction, this number is clearly $a - 1$.

Now let us assume we have trees with a-junctions and b-junctions. The a-junctions have degree $a - 1$, and the b-junctions have degree $b - 1$.

2.2. **Trees with double junctions.** Now let us consider a family of trees with both a-junctions and b-junctions, for a and b two different integers, with the following additional **interchange relation**:



How can we represent this equality topologically? In topology, we can always relax an equality between two objects by creating a path between them. We add an ab-junction which represents this path:



The idea is that if our tree lives in dimension N, we are adding an extra dimension which represents time. The position of t in $[0, 1]$ indicates what the form of the tree is in a continuous way as depicted in figure **??**. Now our tree is parametrized by an element in $[0, 1]^{N+1}$.

In the program, there is a function $make_family$ which holds the combinatorial information about the tree, and a function $tree_at$ which to a point in $[0, 1]^{N+1}$ and

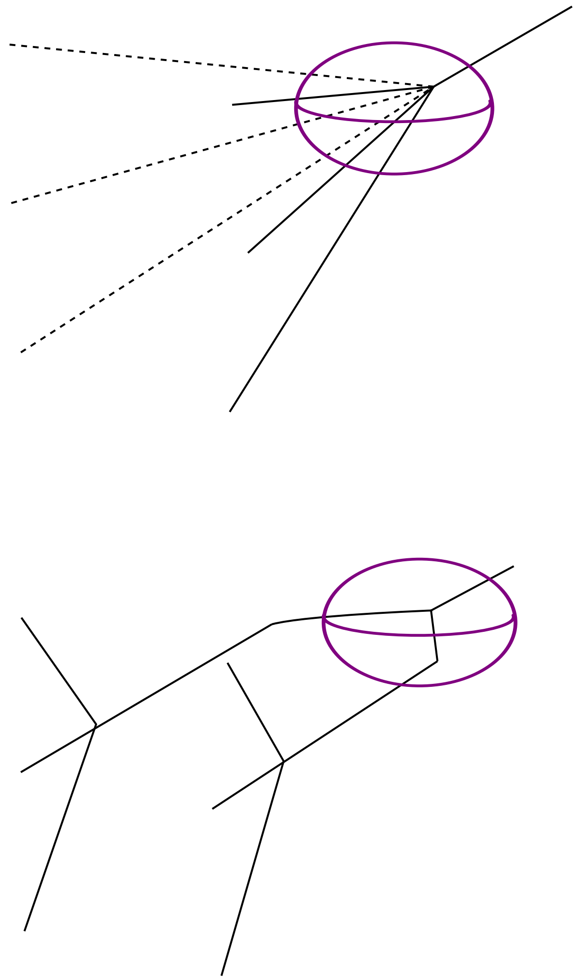


Figure 1. Two trees very close to each other but with very different "local images". The trees are drawn in the direction they are oriented.

the combinatorial information of the tree associates the corresponding embedding of a tree inside a unit cube $[0, 1]^N$.

We would like to determine the degree of this extra junction. It is tempting to say that the degree of this junction is $ab - 1$. However, this does not really work, in part because our definition of degree is not precise enough.

Let's assume we have an ab-junction of this form:

In a neighborhood of this ab-junction however, we have two very different pictures. So if we move just a little bit away from the ab-junction, we can end up in two very different scenarios which have different degrees.

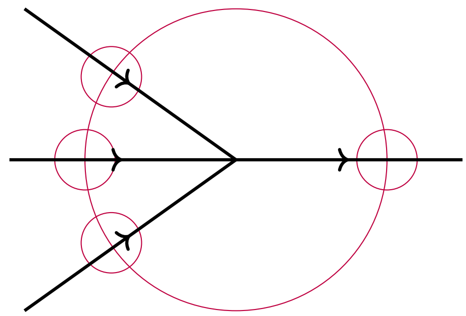


FIGURE 2. The case of a simple junction

2.3. **New degree definition.** Let's give another definition of the degree by going back to the case of a simple junction. If we imagine a smaller lens traveling around the junction and recording what it sees, we will see a single path $a+1$ times: a time with positive orientation and once with negative orientation, and so we recover a degree of $a-1$.

We will find the degree of the ab-junction by traveling around it and recording what we see. Because the ab-junction lives in "higher dimension", we need to count the number of a-junctions and b-junctions that we see around it, not just paths.

Explain this better. Also I should try to make a picture for the bigger junctions but I'm just not sure how, the point is that it's hard

This is the basis of the heuristic we will follow to define the detector map.

3. Defining the detector map

We create a detector a-junctions, and one for b-junctions, in the same way with the following procedure. Given a specific embedding of a tree, we look at a smaller detector cube centered inside the unit cube and we record what it contains.

4. Degree of a map between spheres

The notion of degree we defined in the file tree-problem is closely related to the notion of degree of maps between spheres. Basically, the degree of a map from a sphere S^N to itself is the number of times the map goes around the target sphere, up to homotopy. It is clearer to see this on an example: